

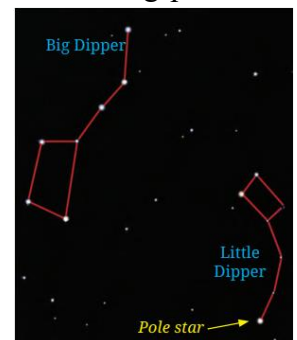
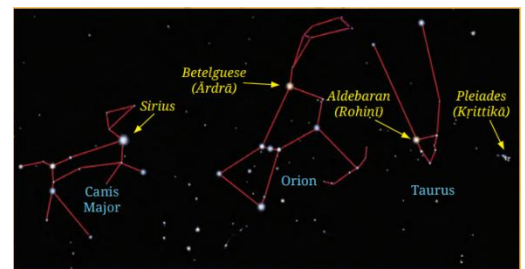
Chapter – 12: Beyond Earth

- Nubra – beautiful region – Ladakh – Youngdol and twin brother Dorjay – live here
- They love – surroundings – mountain peaks, glaciers, night sky – entire sky – 1000s of stars
- Weather – Nubra – almost cloudless – no air, light pollution – night sky – clearly visible
- Yangdol, Dorjay – obs. stars – experience sense of awe
- Growing up – Youngdol, Dorjay – hear stories – stars – from elders
- They heard – particular stars – clear skies – help caravans – find directions
- They wonder – how far, big – these stars
- They enjoy – finding patterns – stars – remind them – similar obj.
- Activity –
 - Following fig. – stars – one part – night sky
 - Look carefully – imagine pattern – grp. of stars
 - Draw lines – connect stars – make pattern
 - Think – obj. – similar to pattern – write your name
 - Repeat same step – more patterns
 - Now think – interesting story – patterns
- Compare patterns – patterns by friends – narrate story – listen stories



Stars and Constellations

- Night – look up – many stars
- Some stars – bright – others – dim – stars – shine – own light
- Some grp. of stars – make pattern – very similar
- Long ago – watching stars – night sky – favorite pastime – our ancestors – identified patterns – animals, obj.
- Many cultures – names – patterns – own stories – these shapes – helped recognizing stars
- Recognizing stars, patterns – useful skill – navigation
- Before modern tech. – before compass – helped people – find directions
- Earlier times – grp. of stars – patterns – constellations
- Currently – regions of sky – include stars – constellations
- BUT – constellations – pattern of stars – often prominent – term constellation – still used – grp. of stars
- Stars – joined – imaginary lines – easy identification
- Orion – represented as hunter – 3 stars – middle – represent belt
- People imagine – hunter Orion – followed by dog (Canis Major) – battling bull (Taurus)
- Canis Major – contain Sirius – brightest star
- 2 distinct patterns – Big Dipper, Little Dipper
- **Pole Star** – Polaris – part – Little – Dipper
- Pole star – appears stationary – North dir. – helps locate



Night Sky Watching

- Clear cloudless night – large no. of stars – visible

- Big stay – sky – not clear – few stars – visible – reason – light pollution, smoke, dust
- Presence – excessive artificial light – night time – light pollution
- Villages, areas – less light pollution – larger no. of stars
- Your house – surrounded – tall buildings, trees – block your view
- Night sky – best viewed – open dark areas
- All stars, constellations – not visible – all places, all nights
- Pole Star – not visible – southern hemisphere
- Identify – particular star, constellation – need to know – how it looks, where to look for it
- Find out – when, which portion of sky – star, constellation – visible – take help – sky mapping apps
- Preparation for night sky watching –
 - Under guidance – adults – identify – dark open areas – away from lights, tall buildings, trees
 - Choose date, time – what you plan to identify
 - Choose moonless night – no clouds – particularly Pole star – not very bright
 - May be useful – access to mobile app – sky map – also carry – compass – find dir. – notebook – draw obs.
 - Selected date, time – go to selected place – sky watching to be done
 - Reach there – wait about half hour – eyes adjust to darkness – helps – see the sky better
- Easily identify – Big Dipper, Pole Star – night sky
- Activity –
 - Look – Big Dipper – summer time – early part of night – around 9 p.m.
 - View sky – above horizon – northern part of sky
 - Identify Big Dipper – now locate – Pole Star
 - Look – 2 stars – end of Big Dipper's cup – imagine – straight line – passing through these 2 – towards North
 - 5 times the dist. – b/w these stars – another star – not very bright – Pole Star
- Also locate – bright constellation – Orion – star – Sirius – night sky
- Activity –
 - India – Orion – best viewed – December to April – after sunset
 - 3 bright stars – short straight line – middle of Orion – belt of hunter – identify 1st – easy to spot
 - Identify Orion – easy to locate – Sirius – close to Orion
 - Imagine – straight line – passing through 3 stars – towards east – lead to Sirius

Our Solar System

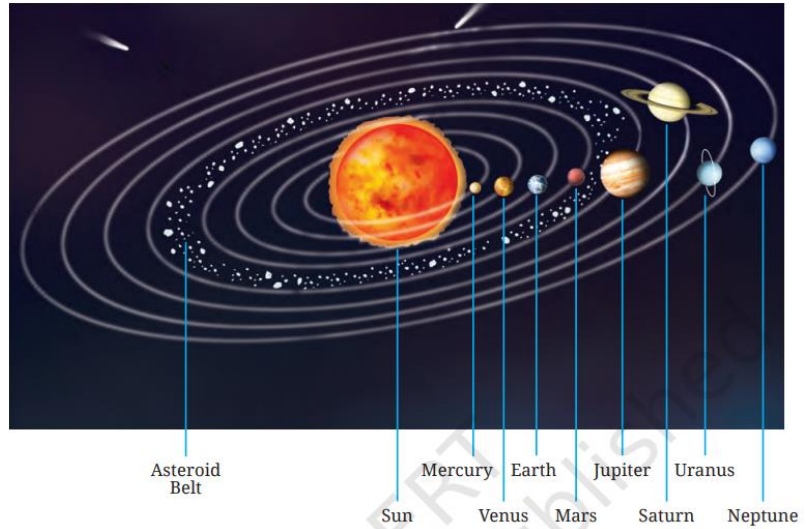
The Sun

- Sun – star – closest to us – extremely hot spherical ball of gases
- Sun – huge amt. of energy – glows brightly
- Sun – produces – heat, light – main src. – energy on earth
- How big – 100x bigger than earth – diameter
- Looks so small – reason – very far from earth
- Dist. b/w sun and earth – 150 million km. (approx.)
- Useful unit – expressing dist. – solar system – astronomical unit (au) – approx. dist. b/w sun and earth
- Most ancient civilisations – Sun – status of deity – worshipped as *Surya*
- Sun – much closer – compared – other stars – appears – much bigger
- Other stars – much farther – appear as points – even though – much bigger than Sun
- Daytime – extreme brightness – Sun – not possible – view – other stars

- Many more obj. – our sky
- Earth – some other obj. – Sun – together form – Solar System
- Most obj. – move around sun – **revolution**

Planets

- Large, nearly spherical obj. – revolve around sun
- Earth – planet – revolves around sun – nearly one yr. – one revolution
- Like earth – other planets – revolve around sun
- While revolving – earth – also rotates – own axis – 24 hrs. – called a day
- 8 planets – increasing order – dist. from sun – Mercury, Venus, Earth, Mars, Jupiter, Saturn, Uranus, Neptune
- Inner 4 planets – smaller in size – nearest to sun – solid surfaces with rocks
- Venus – shine brightly – dawn, dusk – called – morning, evening star – technically – not star
- Mars – called – red planet – appears red – reason – soil on mars – reddish
- Large portion – Earth's surface – covered with water – appears blue – also called – blue planet
- Outer 4 planets – much larger – farther from sun – made of gases
- These giants – ring like structure – made of – dust particles, rocky materials
- Planets – all energy – sun – farther they are – colder they are
- Presence of atm. – trap heat – change temp. – main reason – Venus – hotter than Mercury – BUT – farther from Sun
- Among planets – Venus – easiest to identify – very bright
- After Sun, Moon – Venus – brightest obj.
- Mercury, Mars, Jupiter, Saturn – also seen – naked eye – so far – look like star – shining dots
- Stars – twinkle a lot – planets – do not
- Activity –
 - Most year – Venus – located – dusk, dawn
 - At dawn – look – Eastern dir. – before sunrise
 - At dusk – look – Western dir. – after sunset
- Many obj. – sky – seen – naked eyes – BUT – use telescopes – view better
- Telescope – also helps – view obj. – not visible – naked eyes



Natural satellites

- Obj. – move around planets – satellites – smaller than planet
- Moons – natural satellites
- Earth – 1 Moon – Mars – 2 moons – Jupiter, Saturn, Uranus, Neptune – larger no. of moons

The Moon

- Earth's natural satellite – the Moon – 27 days – complete revolution – around Earth
- Unlike Earth – hardly any atm. – the Moon
- Moon – quarter size – Earth – in diameter
- Moon's surface – bowl-like structure – craters – formed – impact – asteroids, rocks – hitting the surface

- No atm., water, life – Moon's surface – these features – stay long
- Moon – very far – humans – sent spacecraft – explore Moon
- India – also launched – 3 Chandrayaan missions – study Moon

Asteroids

- Sun, planets – nearly spherical – BUT – many small obj. – Solar System – rocky, irregular – asteroids
- Many asteroids – revolve around Sun – path b/w – Mars, Jupiter – this region – asteroid belt
- Occasionally – asteroids pass – close to Earth

Comets

- Sometimes – visitors from outer regions – Solar System – long tails – Comets
- Made of dust, gases, rocks, ice – comets – approach the Sun – frozen material – starts evaporating
- This evaporating material – forms the tail
- Appear dim – move away from Sun – then – not visible – naked eye
- Many comets – revolve around Sun – approach Sun – periodically
- Few comets – escape – move out – Solar System
- Few others – break down and/or fall into – Sun, other planets

- **Sun, 8 planets, their moons, smaller obj., asteroids, comets – together – Solar System**
- Sun, our star – largest, heaviest obj. – Solar System
- Sun – produce – all energy – entire Solar System
- All other obj. – Solar System – shine – reflect sunlight

Milky Way Galaxy

- Moonless night sky – view from dark locations – obs. – extended faint band of light – north to south – across sky
- This – white band – home galaxy – Milky Way (*Akasha Gangah*)
- Galaxy – millions to billions of stars
- Our Solar System – part – Milky Way Galaxy

The Universe

- Many galaxies – outer space – beyond Milky Way
- Scientists – study them – understand stars, galaxies, universe
- We – no idea – life exists – elsewhere
- Search for life – directed mostly – exoplanets – planets – revolving around other stars – our galaxy
- Till now – scientists – not found any life – BUT – still searching